SEVA INNOVATION CHALLENGE 2026 • MAHARASHTRA ZONE

NyaySaakshya

Forensic triage for email-traffic captures. It tells cyber-cell investigators what a capture does, and does not, prove. It does not detect fraud or identify suspects.

Applicant: **Siddhant Hiralal Patil**

Application type: Individual (team size 1)

Sector: Law and Order

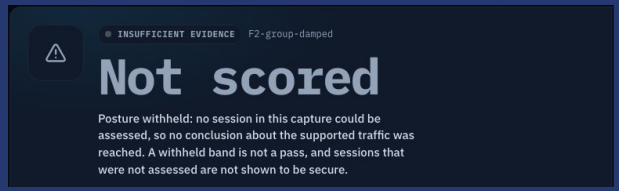
Sub-theme: Cyber-fraud, digital extortion and financial-crime intelligence

Stage: **Working Prototype**

Prototype screens (synthetic lab captures)



Scored: every finding cites its evidence



Withheld: no score when the capture cannot support one

2

The Problem

What real challenge in society, government service or industry are you solving?

What is the problem?

Investigators often hold a network capture that is cut short or encrypted. Reading more into it than it shows can produce a conclusion the evidence cannot support, and weaken a cyber-fraud case.

Who is affected, and where?

Cyber-cell and forensic investigators, and the supervisors who must defend a finding.

Proposed first users: a Maharashtra cyber cell or forensic lab. No partner has agreed yet.

Evidence

- PIB cites over 17.82 lakh CFCFRMS cyber-fraud complaints (7 Oct 2025; period not stated; not email-specific).
- Wireshark decrypts TLS only if keys are supplied (Wireshark docs).
- SMTP security "favors failing open" (Durumeric et al., IMC 2015).

Current gaps

Packet tools show packets; the analyst decides what is safe to conclude. For cut-off or encrypted captures that judgement is manual and hard to repeat or check.

17.82 lakh

complaints cited for the national CFCFRMS cyber-fraud system (PIB, Oct 2025)

Why it matters

Each case may rest on digital evidence that must be read carefully and defended. How many involve email is not published, so we make no claim about it.

3 Our Solution & Innovation

What have you built or designed, how does it work, and what is new about it?

Our solution in brief

NyaySaakshya reads a saved capture of email traffic and labels every fact: observed, inferred, unknown, ambiguous, incomplete or not observable. Each finding points to its packets. If the capture cannot support a score, it gives none.

How it works

1 Input

Investigator uploads a saved email-traffic capture

2 Process

SHA-256 fingerprint; sessions rebuilt; standards-based rules judge only what is visible

3 Output

Cited findings, a score or "not scored", and a report

What makes it new?

	Packet tools alone	NyaySaakshya
What you get	Packets, fields, logs	Cited findings with an evidence state
Cut-off or encrypted	Analyst decides	Marked incomplete or not observable
Repeatable?	Depends on the analyst	Same capture + software, same ID
Role	General analysis	Narrow: evidence-bounded triage

Complements packet tools; no speed or cost comparison is made.

Key USP

Every finding says how sure the evidence allows us to be, and which packets it rests on. When a capture is not enough, it says so instead of guessing.

4 Feasibility & Implementation

How ready is the solution, and what will it take to implement it on the ground?

Current stage: Working Prototype

Idea — Proof of Concept — ✓ Working Prototype — Field Pilot — Ready to Deploy

Technology

Built and running: Python, TShark, React console, FastAPI, SQLite.

Internal validation only: 1,521 automated tests; 98 controlled captures. No external pilot, real-world accuracy benchmark or approval.

Not production-ready. Missing: login, audit log, encrypted storage, backups, a concurrency fix.

Estimated cost

Local synthetic demo: no extra cost.

Hosted prototype: free tiers wipe stored data when idle; Vercel Hobby is $0 for personal use only.

On-premise pilot: not priced. Needs vendor quotes for a server and encrypted storage, plus security engineering.

Estimates only, not quotations.

Timeline

3 months: close the security gaps; tests pass.

6 months: technical pilot on synthetic or consented captures; collect baselines.

12 months: supervised single-unit pilot, only if a department agrees.

Proposed, not committed.

Partners

None yet. Seeking one cyber cell or forensic lab for a supervised pilot. No department has agreed.

5

Impact & Scalability

What change will your solution create, and how can it grow beyond the first location?

1 pilot (target)

Beneficiaries reached

Target: one supervised cyber-cell pilot in year one. None reached yet; no partner yet.

To be measured

Saving in cost / time

Pilot to compare time to a first assessment and to a finished report with current practice. No baseline yet.

Not claimed

Jobs or livelihoods

No evidence of job creation, so none is claimed.

Social impact

Helps investigators avoid unsupported conclusions, supporting fairer and stronger cyber-fraud cases.

Links to SDG 16 (institutions that combat crime, target 16.a).

Social impact is not yet measured.

Sustainability

No revenue model is decided. The stack is open source; licences are to be confirmed.

Option to discuss with a partner: department budget or grant. No commitment exists.

Scale-up plan

Demo, then close the security gaps, then one supervised cyber-cell pilot, then further units.

Each unit would run locally so evidence stays in the unit; central administration only.

Proposed; nothing is deployed.

Pilot measures: time to first assessment, time to report, share of findings with usable provenance, correct handling of unresolved evidence, reproducibility, supervisor confidence.

6

Team/Individual & Support Required

Who is behind the idea, and what help do you need from the Challenge?

SP

Siddhant Hiralal Patil

Applicant and team lead

Individual application (team size 1)

Individual application

No other team member, mentor, partner or funder is named.

Evidence of work to date: 1,521 automated tests and 98 controlled captures (internal validation; see slide 4).

Thank you | Siddhant Hiralal Patil

Support we are seeking

- [x] **Mentorship & technical guidance (security)**
- [x] **Pilot with a government department**
- [x] **Funding / grant support (secure deployment)**
- [ ] Incubation & workspace
- [x] **Testing, certification or IP support**

With security mentorship and one supervised pilot, we can move from a tested prototype to a measured trial of whether NyaySaakshya helps investigators avoid unsupported conclusions.